

Problem 1 (13 marks)

Given: Home $(K, L) = (300, 900)$ so $(K/L)_H = 300/900 = 0.33$.

Foreign $(K, L) = (800, 800)$ so $(K/L)_F = 800/800 = 1.00$.

T is labour-intensive; S is capital-intensive. Identical technologies; homothetic preferences; perfect competition; full employment.

- (1) **Relative factor abundance.** A country is *labour-abundant* if it has a lower K/L than its trading partner. Since $0.33 < 1.00$, Home is labour-abundant and Foreign is capital-abundant.
- (2) **Trade pattern (H–O theorem).** With identical technologies and homothetic preferences, each country exports the good that uses its abundant factor intensively. Therefore, Home (labour-abundant) exports T ; Foreign (capital-abundant) exports S .
- (3) **Factor prices (Stolper–Samuelson).** Opening to trade raises the relative price of the export good in each country (versus autarky). Hence in Home, the relative price of T rises $\Rightarrow$ the *real wage* (return to labour, the intensive factor in T) **rises** and the *real rental rate* **falls**. In Foreign, the relative price of S rises $\Rightarrow$ the *real rental* **rises** and the *real wage* **falls**. (Results hold in real terms with respect to both goods under the usual assumptions.)
- (4) **A 10% rise in P_T/P_S (Home).** By the magnification effect, the percentage change in factor prices exceeds the percentage change in the price of the good that uses that factor intensively. Thus the *wage* rises by $>10\%$ (in real terms), while the *rental* falls (in real terms). Sectoral outputs adjust in the long run: more resources flow to T and away from S (Rybczynski logic at given prices; with the price change, T expands and S contracts).
- (5) **Mechanism for factor price equalization (FPE).** With identical technologies, common goods prices under free trade pin down unit-cost equations:

$$P_T = a_{LT}w + a_{KT}r,$$

$$P_S = a_{LS}w + a_{KS}r.$$

Given (P_T, P_S) and strictly concave, competitive technology, (w, r) is determined uniquely. If both countries produce both goods (diversification) and share technologies, they face the same (P_T, P_S) and hence converge to the same (w, r) in the *cone of diversification* (the FPE set).

Problem 2 (13 marks)

Given: PPF $A^2 + 2E^2 = 900$. World relative price $P_E/P_A = 3$. Preferences satisfy $E = 0.5 A$ at the consumption point.

Normalization. Set $P_A = 1 \Rightarrow P_E = 3$. National income at production point: $V = A + 3E$.

(1) Value-maximizing production (A_P, E_P)

Maximize $V = A + 3E$ subject to $A^2 + 2E^2 = 900$. At an interior optimum, the slope of the PPF equals the slope of the isovalue line:

$$\text{PPF: } A^2 + 2E^2 = 900 \Rightarrow 2A dA + 4E dE = 0 \Rightarrow \frac{dE}{dA} = -\frac{A}{2E}.$$

$$\text{Isovalue slope} = -\frac{P_A}{P_E} = -\frac{1}{3}.$$

Tangency gives $-\frac{A}{2E} = -\frac{1}{3} \Rightarrow \frac{A}{2E} = \frac{1}{3} \Rightarrow 3A = 2E \Rightarrow E = \frac{3}{2}A$.

Substitute into PPF:

$$A^2 + 2\left(\frac{3}{2}A\right)^2 = 900 \Rightarrow A^2 + 2 \cdot \frac{9}{4}A^2 = \left(1 + \frac{9}{2}\right)A^2 = \frac{11}{2}A^2 = 900.$$

Hence

$$A_P^2 = \frac{1800}{11}, \quad A_P = \sqrt{\frac{1800}{11}}, \quad E_P = \frac{3}{2}A_P.$$

It is often convenient to keep the exact form. Numerically,

$$A_P \approx 12.788, \quad E_P \approx 19.182.$$

(2) Consumption (A_C, E_C) , export good, and trade values

Production value (national income) at world prices:

$$V = A_P + 3E_P = A_P + 3\left(\frac{3}{2}A_P\right) = \frac{11}{2}A_P.$$

Budget line at world prices: $A + 3E = V$. Preferences require $E = 0.5 A$. Substitute into budget:

$$A + 3(0.5A) = 2.5A = V \Rightarrow A_C = \frac{V}{2.5} = \frac{\frac{11}{2}A_P}{\frac{5}{2}} = \frac{11}{5}A_P, \quad E_C = 0.5 A_C = \frac{11}{10}A_P.$$

Thus

$$A_C = \frac{11}{5}A_P, \quad E_C = \frac{11}{10}A_P.$$

Trade pattern (compare P vs. C):

$$\text{Exports of } E : X_E = E_P - E_C = \left(\frac{3}{2} - \frac{11}{10} \right) A_P = \frac{4}{10} A_P = \frac{2}{5} A_P > 0,$$

$$\text{Imports of } A : M_A = A_C - A_P = \left(\frac{11}{5} - 1 \right) A_P = \frac{6}{5} A_P > 0.$$

So the country **exports electronics** and **imports agricultural products**.

Values at world prices:

$$\text{Export value} = P_E X_E = 3 \cdot \frac{2}{5} A_P = \frac{6}{5} A_P, \quad \text{Import value} = P_A M_A = 1 \cdot \frac{6}{5} A_P = \frac{6}{5} A_P.$$

(3) Balanced trade verification

Export and import values are equal at world prices ($\frac{6}{5} A_P$), so trade is balanced:

$$P_E X_E = P_A M_A.$$

(4) Effect of an increase in P_E/P_A on production, consumption, and welfare

An increase in P_E/P_A (steeper world price line) implies:

- **Production:** Tangency condition $-\frac{A}{2E} = -\frac{P_A}{P_E}$ gives a more negative slope on the right-hand side $\Rightarrow$ production tilts toward E (higher E/A).
- **Consumption:** The trade (budget) line through the new production point pivots outward in E -intercepts; with homothetic preferences, consumption shifts toward more E as well.
- **Welfare:** Because the country is an E -exporter, a rise in P_E/P_A improves its *terms of trade*. With no distortions, the budget set expands and the country reaches a higher indifference curve. **Welfare rises.**

(5) Export-biased growth concentrated in electronics

- **Relative supply:** The PPF rotates outward more in the E direction; at given prices, $RS = Q_E/Q_A$ rises (rightward shift).
- **Large-country effect on prices:** If the country is large, the world relative price P_E/P_A tends to *fall* (worsening the country's terms of trade) because its export good becomes more abundant globally.
- **Welfare:** Two opposing effects — (i) *production gain* from a larger PPF; (ii) *terms-of-trade loss*. Typically welfare still rises; only in extreme cases with very inelastic foreign demand could the TOT deterioration outweigh the production gain (“immiserizing growth”).

Numerical illustration (optional). Using $A_P \approx 12.788$:

$$E_P \approx 19.182, \quad V = \frac{11}{2}A_P \approx 70.334, \quad A_C = \frac{11}{5}A_P \approx 28.134, \quad E_C = \frac{11}{10}A_P \approx 14.067.$$

Exports: $X_E \approx 5.115$ with value $3X_E \approx 15.345$; Imports: $M_A \approx 15.346$ with value ≈ 15.346 (balanced up to rounding).